Savneet Bal

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Education

McMaster University Hamilton, ON
Bachelor of Engineering Co-op (BEng), Computer Engineering & Management *2024 - Expected 2029*

- **GPA:** 3.8/4.0 | Dean's Honour List
- **Relevant Coursework:** Digital Logic Design (Verilog, FPGA labs), Circuit Analysis (PSpice, oscilloscopes), Programming & Data Structures, Engineering Design & Prototyping

Experience

Assistant, Creative Technology & Digital Literacy Programs May 2026 – Aug 2026
Caledon Public Library *Caledon, ON*

- Maintained 3D printers, laser cutters, Cricut and embroidery machines, scanners, and printers, minimizing downtime
- Diagnosed hardware and software faults by independently isolating root causes and testing fixes, supporting 100+ patrons
- Taught patrons to operate laser cutters and 3D printers safely by following equipment procedures
- Facilitated 20+ STEM, maker, and technology programs for 70+ participants of varying experience levels

Summer Coding Camp CIA Student Leader 2023 – 2024
Peel District School Board *Brampton, ON*

- Led coding workshops for 40+ students, teaching programming, app development, and problem solving
- Mentored students in troubleshooting, debugging, and logic design, helping 40+ students complete their own projects
- Collaborated with fellow leaders and instructors to plan lessons and adjust pacing to students' skill levels

iOS Developer Intern Jun 2022 – Aug 2022
Our Wave Hub x Apple Inc *Brampton, ON*

- Developed a SwiftUI iOS application from concept to implementation in a cross-functional agile team
- Resolved 10+ bugs by reproducing issues, implementing fixes, and validating them through iterative testing
- Communicated progress and blockers to teammates and mentors to keep work aligned with requirements

Projects

Automated Dispensing Cabinet (ADC) Software System | *Python, UML, Git* Sept 2026 – Present

- Specified 44 requirements and modelled a safety-critical medication cabinet with UML state machine diagrams
- Designed a modular architecture and implemented dispensing, inventory, logging, and hardware fault handling in Python
- Performed an FMEA on hardware/software failure modes (e.g., drawer faults) to define safety requirements
- Wrote test plans traced to safety requirements and defined the interface to a partner team's Monitoring System

Smart Walk AI - Power in Community | *Raspberry Pi, Python, AutoCAD* Jan 2025 – Apr 2025

- Integrated a Raspberry Pi, camera, and Bluetooth audio module into a portable real-time object detection system
- Optimized power consumption to achieve 8+ hours of operation and validated reliability through iterative testing
- Designed and prototyped an ergonomic AutoCAD enclosure, optimizing component placement and durability
- Collaborated across hardware, software, and enclosure design to deliver a working prototype by the deadline

Luggage Handling Automation System | *Python, Q-Arm Robotics, AutoCAD, 3D Printing* Oct 2024 – Dec 2024

- Developed a robotic luggage transfer system using a Q-Arm, rack-and-pinion mechanism, and custom 3D-printed fixtures
- Debugged control errors through iterative testing, improving system stability and reliability
- Designed custom components in CAD and integrated mechanical and software subsystems into one working prototype
- Divided mechanical, control, and software tasks among teammates and coordinated integration testing

Skills

Digital Design & Electronics: FPGA development (Quartus), Verilog, digital logic, combinational circuit design, circuit analysis, PSpice simulation, soldering, oscilloscopes, function generators, multimeters

Hardware & Embedded: Raspberry Pi, Q-Arm robotics, hardware/software integration, prototyping

Programming: Python, C/C++, Java, MATLAB, Swift (SwiftUI)

Engineering Practices: FMEA, requirements specification, UML modelling, test planning, root-cause analysis, debugging, agile workflows

CAD, Fabrication & Tools: Autodesk Inventor, AutoCAD, 3D printing, laser cutting, Git, basic Linux, VS Code